

Volume V · Chapter 26 — LLMs/VLA as Planners & Human-in-the-Loop Safety · Labs

Source: split from med-tech-curriculum-V1.md (V1). From this split onward this chapter is maintained **independently** here. See Volume-V-Chapter-26-context.md for the AI interaction log.

Prerequisites: Ch 11, Ch 25.

Learning outcomes — the student can: use LLMs/VLA models as high-level planners; translate NL tasks into verified action sequences; design human-in-the-loop approval; describe VLA SOTA and its limits in medicine.

Labs (hands-on) — 12 h:

#	Lab	h
26a	LLM task planner: NL instruction → action plan in sim	4
26b	Add verification/validation of generated plans	4
26c	Human-approval gate before execution; log & audit	4

Datasets/tools: local LLM; the simulator; a planning framework. **Assessment:** LLM-planner-in-sim with approval gate (**60%**); safety analysis (**20%**); quiz (**20%**). **Key decisions:** autonomy level; verification strategy; **when human approval is mandatory**. **References:** plan §7; §13 → *Robotics & embodied AI; Medical LLMs*. **Hours:** Theory **8** + Lab **12** = **20**.